WORKSHEET

Chapter 10: Population parameters

1 The mean of a sample of 40 was 35, with a standard deviation of 7. Determine a 96% confidence interval for the population mean.

2 The mean of a sample of 60 was 115, with a standard deviation of 22. Determine a 90% confidence interval for the population mean.

3 The mean of a sample of 75 was 26, with a variance of 9. Determine a 98% confidence interval for the population mean.

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4 The mean of a sample of 90 was 120, with a standard deviation of 28. Determine a 87% confidence interval for the population mean.

5 The mean of a sample of 80 was 2.5, with a standard deviation of 0.8. Determine a 99% confidence interval for the population mean.

6 The mean of a sample of 50 was 26, with a variance of 4. Determine a 99.9% confidence interval for the population mean.

7 A random sample of 40 households was undertaken in a suburb of Sydney. The weekly household incomes ($) are given below.

1262, 1308, 1061, 1226, 1131, 1218, 1770, 1383, 1816, 1037,
1726, 1334, 1449, 1439, 1178, 1615, 1177, 1249, 1257, 995,
1011, 1555, 1163, 2031, 1898, 1189, 1161, 1358, 1090, 1503,
1040, 1234, 1192, 1261, 1671, 1550, 897, 1498, 1880, 1246,

a Use these results to estimate the mean and standard deviation of Australian households.

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b State the 95% confidence interval for the mean.

c State any problems with the estimates.

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Answers

1 About 32.73 and 37.27

2 About 110.33 and 119.67

3 About 25.19 and 26.81

4 About 115.53 and 124.47

5 About 2.27 and 2.73

6 About 25.07 and 26.93

7 **a** Mean is about $1351.51, standard deviation is about $274.49

b About $1266.44 and $1436.58

c The suburb sampled or the sample taken could be atypical of the Australian population.

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